

2004~2005 学年第一学期《高等数学》期末考试试题 B 卷答案

一、填空题 (4×4 分)

1、 $x^2 - 3$; 2、 $-\frac{1}{3}$; 3、4; 4、 $xf(x) + c$; 5、 $\frac{2x \sin x^2}{1 + \cos^2 x^2}$

二、单项选择题 (5×3 分)

1、C; 2、D; 3、A; 4、C; 5、B

三、试解下列各题

解: 1、 $\lim_{x \rightarrow 0} \frac{1}{x} (\cot x - \frac{1}{x}) = \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x} = \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{-\sin x}{3x^2} = -\frac{1}{3}$

2、 $\lim_{x \rightarrow 0} (1 + 3x)^{\frac{2}{\sin x}} = e^{\lim_{x \rightarrow 0} \frac{2 \ln(1+3x)}{\sin x}} = e^{\lim_{x \rightarrow 0} \frac{6x}{x}} = e^6$

3、 $dy = \left[\arctan x + \frac{x}{1+x^2} - \frac{x}{1+x^2} \right] dx = \arctan x dx$

4、两边对 x 求导

$$e^{x+y} \left(1 + \frac{dy}{dx}\right) - y - x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{e^{x+y} - y}{x - e^{x+y}}$$

5、 $dx = -2t \sin t^2 dt$

$$dy = (\cos t^2 - 2t^2 \sin t^2 - \cos t^2) dt = -2t^2 \sin t^2 dt$$

$$\frac{dy}{dx} = \frac{2t^2 \sin t^2}{2t \sin t^2} = t$$

$$d \frac{dy}{dx} = dt \quad \frac{d^2 y}{dx^2} = -\frac{1}{2t \sin t^2}$$

6、 $\int \frac{1}{\sqrt{x}(1+x)} dx = 2 \int \frac{1}{1+(\sqrt{x})^2} d\sqrt{x} = 2 \arctan \sqrt{x} + c$

7、 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin^2 x}{1+e^{-x}} dx = \int_{-\frac{\pi}{4}}^0 \frac{\sin^2 x}{1+e^{-x}} dx + \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1+e^{-x}} dx$

$$\int_{-\frac{\pi}{4}}^0 \frac{\sin^2 x}{1+e^{-x}} dx \stackrel{\text{令 } x=-t}{=} \int_0^{\frac{\pi}{4}} \frac{\sin^2 t}{1+e^t} dt$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin^2 x}{1+e^{-x}} dx = \int_0^{\frac{\pi}{4}} \sin^2 x dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \Big|_0^{\frac{\pi}{4}} = \frac{1}{8} (\pi - 2)$$

8、 $\int_0^1 \arctan(1 + \sqrt{x}) dx \stackrel{\text{令 } \sqrt{x}+1=t}{=} \int_1^2 \arctan t d(t-1)^2$

$$= (t-1)^2 \arctan t \Big|_1^2 - \int_1^2 \left(1 - \frac{2t}{1+t^2}\right) dt$$

$$= \arctan 2 + \frac{5}{2} - 1$$

四、解：例如广义积分 $\int_0^1 \frac{1}{\sqrt{x}} dx$ 收敛时，但广义积分 $\int_0^1 \frac{1}{x} dx$ 发散。

五、解：令 $f(x) = xe^x - e^x + 1$ ($x > 0$)

$$f'(x) = e^x + xe^x - e^x = xe^x > 0$$

$$\therefore f(x) \uparrow \quad (x > 0)$$

$$\lim_{x \rightarrow 0^+} f(x) = 0 \quad f(x) > 0$$

$$\text{即 } e^x - 1 < xe^x \quad (x > 0)$$

六、证：由题设知： $f(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{6}f'''(\xi)x^3$

$$= \frac{1}{2}f''(0)x^3 + \frac{1}{6}f'''(\xi)x^3, \quad \xi \text{ 在 } 0 \text{ 与 } x \text{ 之间。}$$

$$\text{令 } x = -1, \text{ 得 } 1 = f(1) = \frac{1}{2}f''(0) + \frac{1}{6}f'''(\xi_1) \quad (0 < \xi_1 < 1),$$

$$0 = f(-1) = \frac{1}{2}f''(0) - \frac{1}{6}f'''(\xi_2), \quad (-1 < \xi_2 < 0),$$

$$\text{两式相减得 } 6 = f'''(\xi_1) + f'''(\xi_2) \leq 2 \max\{f'''(\xi_1), f'''(\xi_2)\},$$

$$\text{当 } f'''(\xi_1) \geq f'''(\xi_2), \text{ 取 } \eta = \xi_1, \text{ 有 } f'''(\eta) \geq 3;$$

$$\text{当 } f'''(\xi_2) \geq f'''(\xi_1), \text{ 取 } \eta = \xi_2, \text{ 有 } f'''(\eta) \geq 3.$$

$$\text{故存在 } \eta \in (-1, 1), \text{ 使 } f'''(\eta) \geq 3.$$

七、证：由于 $x_1, x_2 \in (-\infty, +\infty)$ 时恒有

$$|\sin x_1 - \sin x_2| = \left| 2 \cos \frac{x_1 + x_2}{2} \sin \frac{x_1 - x_2}{2} \right| \leq 2 \left| \sin \frac{x_1 - x_2}{2} \right| \leq 2 \left| \frac{x_1 - x_2}{2} \right| = |x_1 - x_2|$$

所以，对任给的 $\varepsilon > 0$ ，取 $\delta = \varepsilon$ ，那么对一切 $x_1, x_2 \in (-\infty, +\infty)$ ，只要 $|x_1 - x_2| < \delta$ ，就有

$$|\sin x_1 - \sin x_2| < \varepsilon \quad \text{故 } f(x) = \sin x \text{ 在 } (-\infty, +\infty) \text{ 上一致连续。}$$

八、解： $s(t) = s_1(t) + s_2(t)$

$$= \int_0^t (t^2 - x^2) dx + \int_t^1 (x^2 - t^2) dx = \frac{4}{3}t^3 - t^2 + \frac{1}{3}$$

$$s'(t) = 4t^2 - 2t = 0, \quad t_1 = 0, \quad t_2 = \frac{1}{2}$$

$$s(0) = \frac{1}{3}, \quad s\left(\frac{1}{2}\right) = \frac{1}{4}, \quad s(1) = \frac{2}{3}$$

$$\text{当 } t = 1 \text{ 时, } s_1 + s_2 \text{ 取最大面积 } \frac{2}{3};$$

$$t = \frac{1}{2} \text{ 时, } s_1 + s_2 \text{ 取最小面积 } \frac{1}{4};$$